

The REN Switchback

Urban Design for the Centennial Jing-Zhang AI Innovation Belt, Pivoting on 人

In 1909 at Qinglongqiao, the Jing-Zhang Railway used a switchback shaped like the character 人 to turn one unclimbable grade into two climbable ones. This proposal turns that wisdom into urban institution: one heritage-park slow-mobility spine, one origin pivot, three climbing segments (人–从–众), two coordinated wings, and twelve REN-clasp stitching crossings — every climb of intelligence passes a reversible, reviewable, roll-back-able REN pivot.

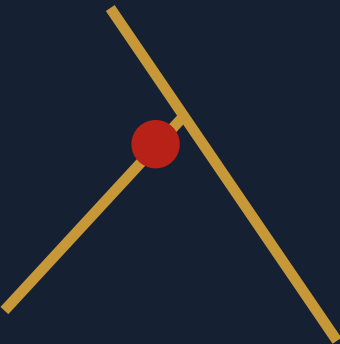
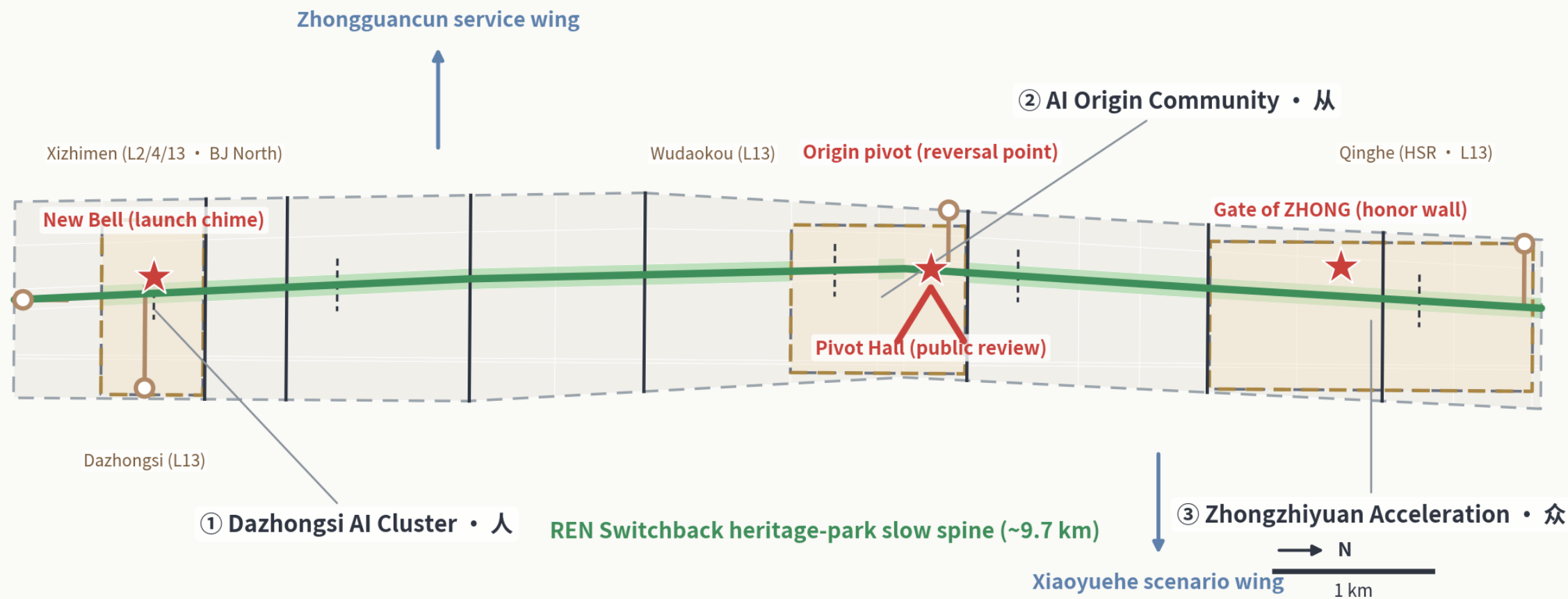


Fig.1 Overall concept, evidence chain, and belt structure

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Fig.1 • Concept: The REN Switchback — an AI belt pivoting on the human

One spine • One pivot • Three grades • Two wings • Twelve stitches



In 1909 the Jing-Zhang Railway conquered an impossible grade at Qinglongqiao with a reverse switchback shaped like 人 — the character for “human” .
One steep line became two climbable ones; the pivot is the room engineering leaves for people.
This proposal translates that wisdom into the AI belt: every climb of intelligence passes a human pivot — reversible, reviewable, revocable.

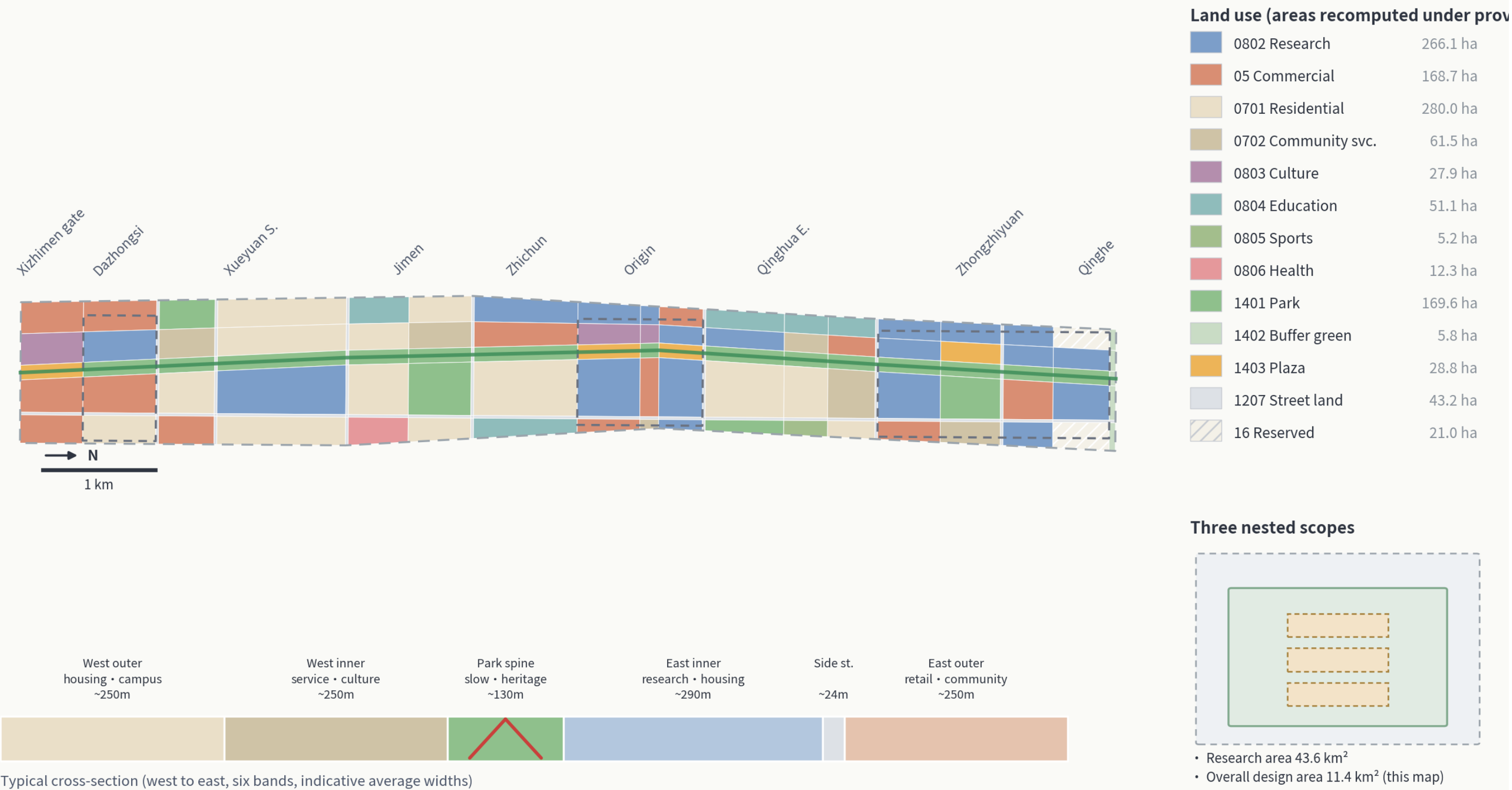
Naming system: 人 REN → 从 CONG → 众 ZHONG (one pivots • pairs collaborate • crowds accelerate)



Boundary status: provisional inference • NOT an official redline • areas recomputed in EPSG:4548 • full recomputation once official data is released
Data: geometry/*.geojson and metrics.json in this package (officially announced areas labelled separately)

Fig.2 • Spatial structure and land use: a mixed innovation belt on a green spine

150 seamless land-use cells • six parallel bands • statutory controls pending



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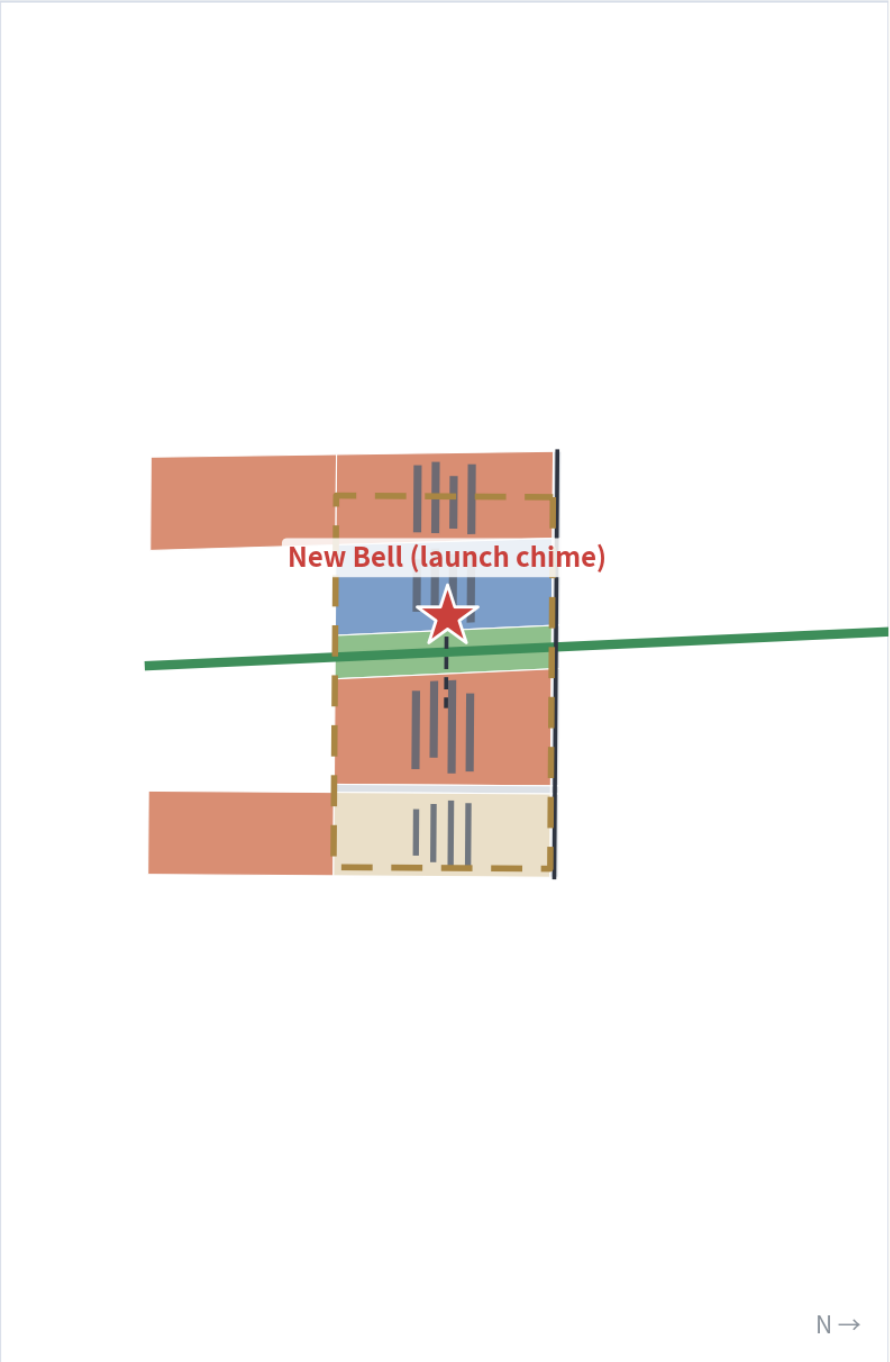
Fig.3 Detailed design index of the three key areas

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Fig.3 • Three key areas in detail: REN • CONG • ZHONG

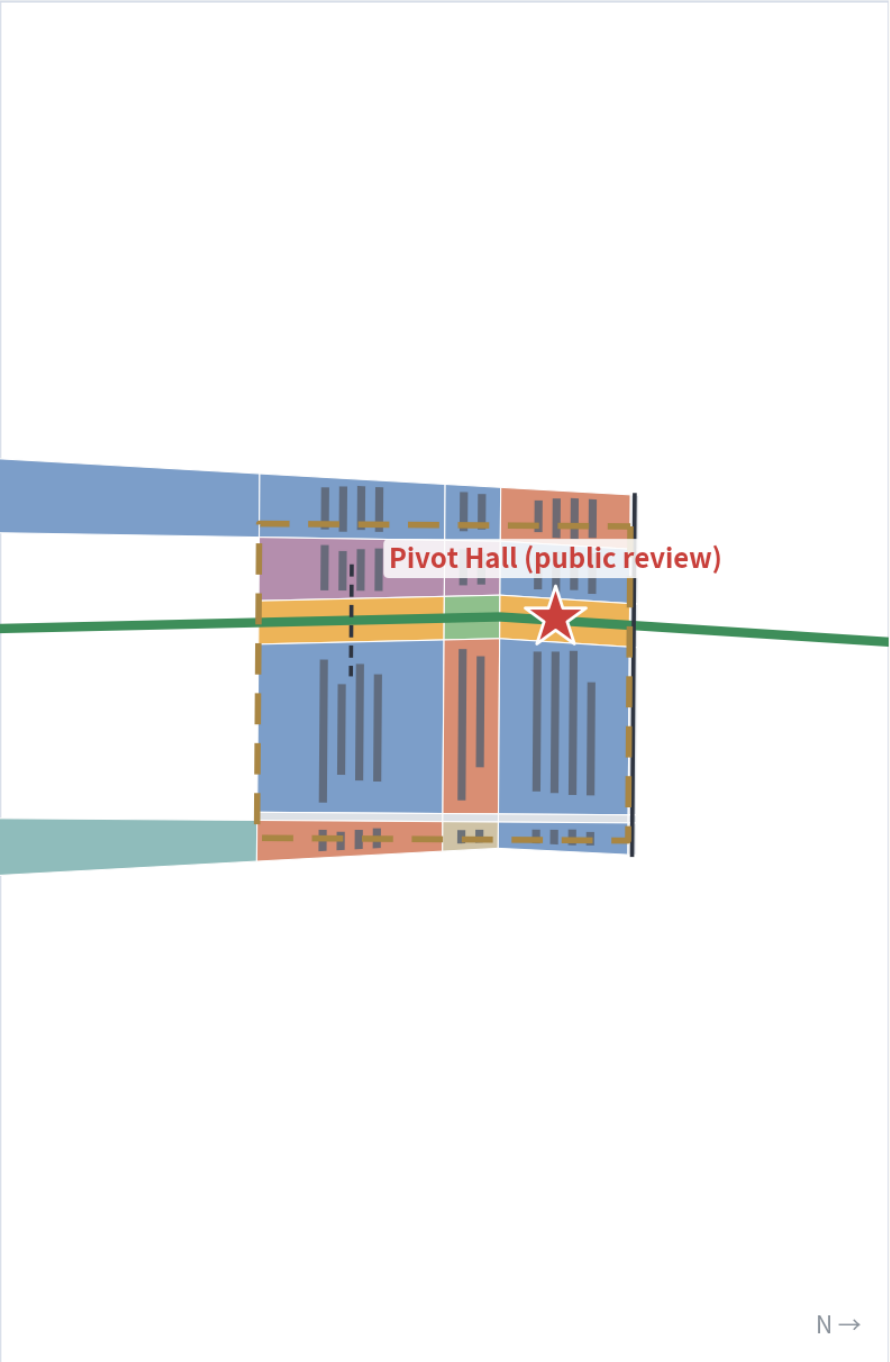
Three ways of climbing on one spine (provisional rectangles; official polygons pending)

① Dazhongsi • 人
AI-native retail & business renewal



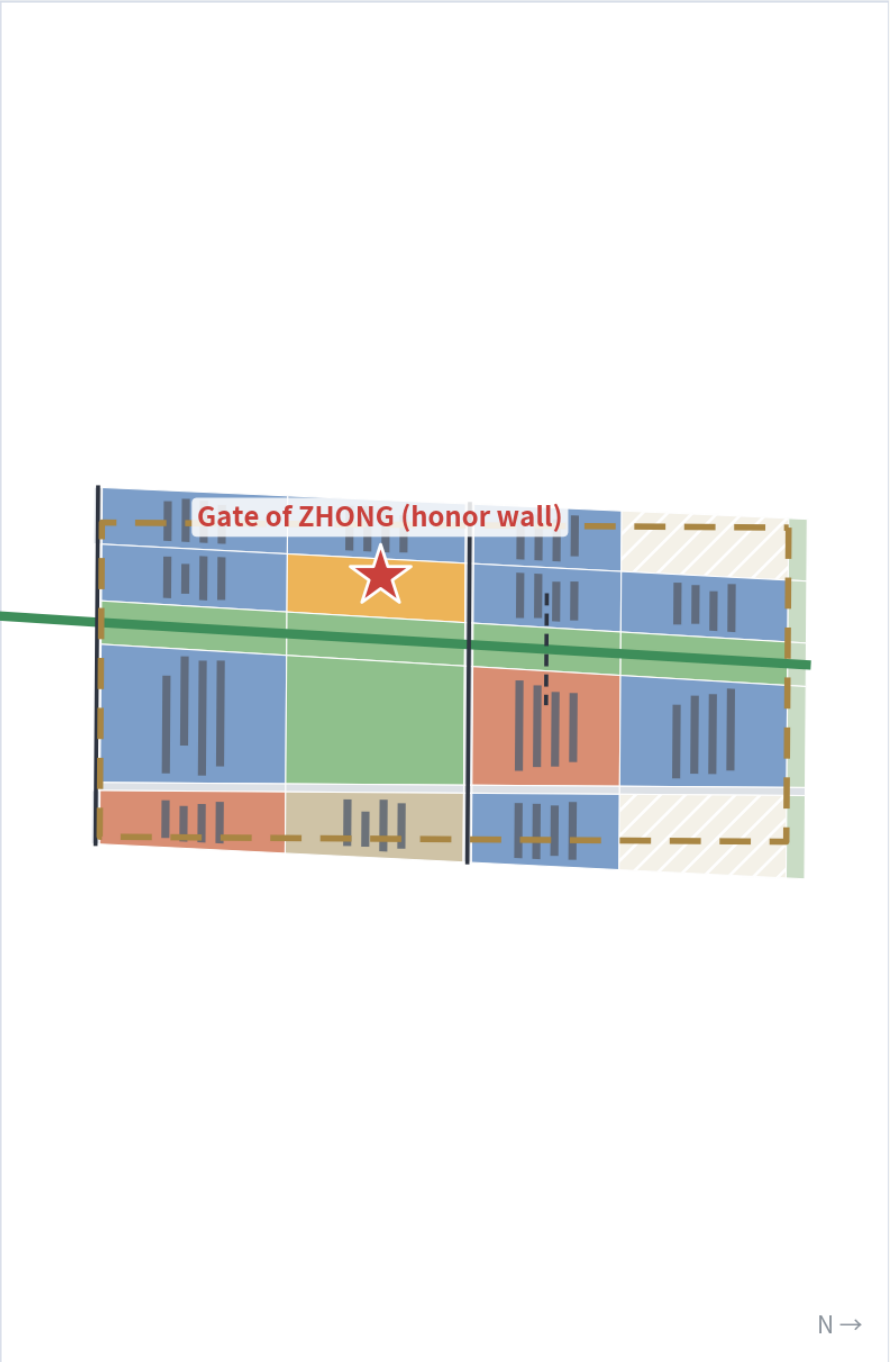
provisional 72.0 ha (announced ≈ 72.0 ha)
launch bell • station stitching • adaptive reuse
Boundary status: provisional inference • NOT an official redline • areas recomputed in EPSG:4548 • full recomputation once official data is released
Data: geometry/*.geojson and metrics.json in this package (officially announced areas labelled separately)

② Origin Community • 从
living room of human-AI collaboration



provisional 104.3 ha (announced ≈ 104.3 ha)
station narrative • public review • open source

③ Zhongzhiyuan • 众
full-stack acceleration ground

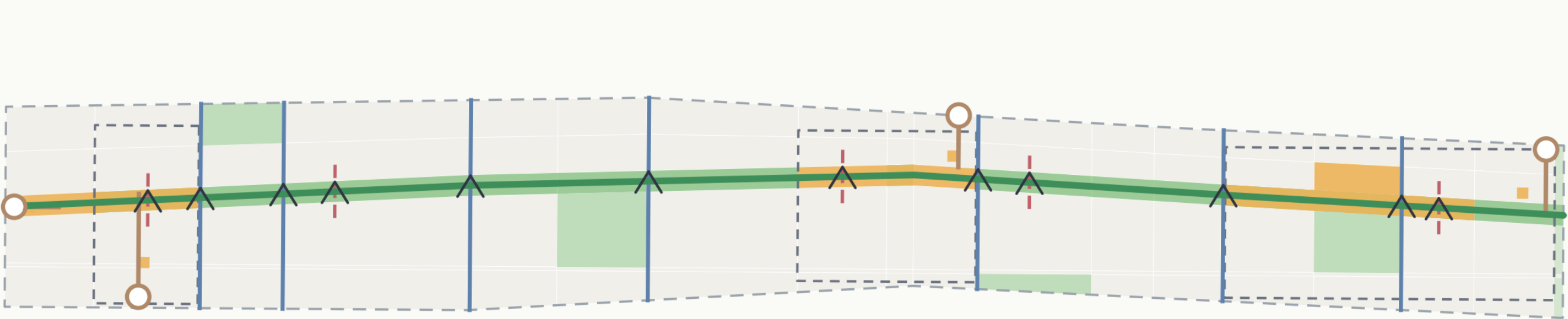


provisional 192.9 ha (announced ≈ 192.1 ha)
research clusters • test scenarios • Qinghe reserve

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Fig.4 • Mobility and blue-green space: twelve stitches

Re-stitching the two sides a century of rail once cut apart



Xiaoyuehe blue-green corridor (east wing)

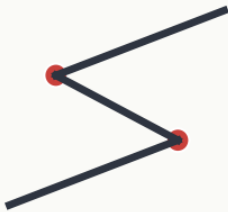
- Vehicular stitch (existing street ref.)
- - Pedestrian stitch (new concept)
- Slow spine (heritage park greenway)
- Transit link

12 stitches

4 transit links

slow spine ~9.7 km

green ratio 15.4%



Standard stitch section: switchback access
(slope $\leq 1:12$ • two-way wheelchair passage)

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Fig.5 Core metrics recomputation and evidence chain

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Fig.5 • Metrics and evidence chain: every number recomputable, every gap declared

GeoJSON → EPSG:4548 → metrics.json → drawings/HTML → four self-check gates

Recomputed metrics (provisional basis)

Site area (recomputed)
1,141.3 ha
announced ≈ 1,140 ha

Green ratio (concept)
15.4%
park spine + pockets + buffer

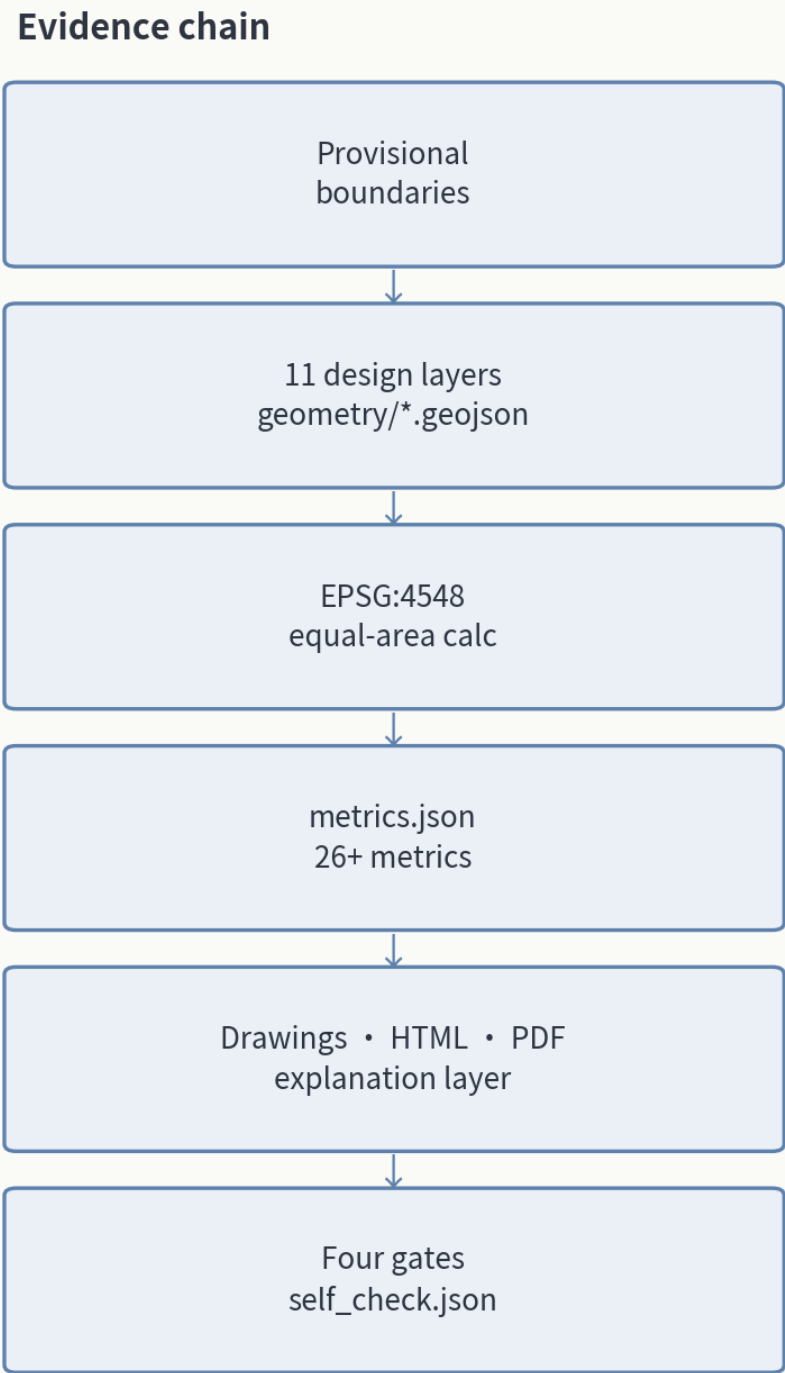
Public space ratio
5.4%
plazas + active park + forecourts

Street-land ratio
3.8%
stitch corridors (not redlines)

Concept building volume
6.46 M m²
low-confidence design massing

FAR (statutory)
pending
no approved controls — not invented

Height (statutory)
pending
aviation/heritage limits unpublished



Pending official data

☐ Official polygons for all three scopes and key areas

☐ Statutory controls (FAR / height / density / green / setback)

☐ Existing building stock, tenure, retain-renovate data

☐ Road redlines, station footprints, municipal capacity

☐ Heritage protection zone GIS layers

Once official data arrives: the whole package is regenerated —
no single boundary file is swapped in isolation.

Coverage: announcement 1.3/1.4/1.5 + agent.1–6 = 23 items → compliance_1

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Core Recomputed Metrics

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Three AI Pilgrimage Landmarks (south to north)

- New Voice of the Great Bell (Go-Live Bell Platform) — every public AI service tolls its go-live and exit; citizens can hear it.
- REN Pivot Station (Public Review Hall) — purpose, data boundary, review records, and exit mechanism of every scenario, open to consultation.
- Gate of the Many (Contributor Honor Wall) — community-adopted contributions engraved in real time, including every participant of this open call.

Gradient-Graded Governance (G1/G2/G3)

- G1 instant human takeover: 8 everyday scenarios — guides, service desks, health kiosks.
- G2 periodic review + one-touch degradation: demand-responsive lighting, elder companionship.
- G3 closed test segments: low-speed shuttle, robot delivery, shared test field — the 3 industry testing scenarios.
- Every scenario card carries a REN fallback plan: when to degrade to human, how to exit, who reviews.

Three Phases (conceptual)

- Near 2026–2028, Through-Running and Launch: spine opened, 12 REN clasps, pivot plaza and Public Review Hall (~383.7 ha).
- Mid 2029–2031, Acceleration and Transformation: Zhongzhiyuan clusters, Dazhongsi consumption transformation, two test segments (~293.6 ha).
- Long 2032–2035, Networking and Synergy: housing-service band renewal, wing interfaces matured, reserves activated (~464.0 ha).

Core Recomputed Metrics

Overall Design Area (recomputed)	1,141.3 ha	announced ~1,140 ha
Three key areas total (provisional)	369.3 ha	announced 368.4 ha
Green space ratio (conceptual)	15.4%	not statutory
Public space share	5.3%	all against the spine
Road land share	3.8%	few roads, many stitches
Conceptual building volume	~6.46M m ²	149 masses, low confidence
Clasps / scenario nodes / landmarks	12 / 15 / 3	tallied from layers
Statutory FAR / height	unknown	await regulatory conditions

All metrics recomputed from this package's GeoJSON in EPSG:4548; statutory FAR and height await official regulatory conditions.

This is a conceptual recommendation and reference scheme for an open call, offered for professional deepening. All spatial layers are generated from a provisional inferred boundary — not the official redline. It constitutes no statutory planning conclusion, government-approved opinion, or implementation commitment. Structured data (GeoJSON/JSON) is authoritative; PDFs are presentation only.